HW1: Due 11:59pm (MyCourses/Assignments/, Thursday, 1/22

Stat-325 Design of Experiments
Worksheet 1 - Probability & Expectation

Problem #1

Consider the random variable X with PDF

$$p(x) = \begin{cases} \dfrac{x}{10}, & x = 1, 2, 3, 4 \\ 0, & \text{otherwise} \end{cases}$$

A. What is $P(X = 1)$? $P(X = 2)$, $P(X = 3)$, $P(X = 4)$?

B. What is the value of $E(X)$?

C. What is the value of $V(X)$?

D. If X_1 and X_2 are independent random variables, each with the PDF $p(x)$, what is the value of $E(4X_1 - 3X_2)$?

E. What is the value of $V(4X_1 - 3X_2)$?

Problem #2

Consider the random variable Y with PDF

$$f(y) = \begin{cases} \dfrac{0.72135}{y}, & 1 \le y \le 4 \\ 0, & \text{otherwise} \end{cases}$$

A. What is the value of f(2)? What is $P(1 \le Y \le 2)$?

B. What is the value of E(Y)?

C. What is the value of V(Y)?

D. What is the value of E(2Y + 1)?

E. What is the value of V(2Y + 1)?